AD 502 – Machine Learning Lab

Pre-Reading Material – Lecture 1
Introduction to Pandas Library

Estimated Reading Time: 25–30 minutes

Reference Reading: Daniel Bourke, 'A Quick Introduction to Data Analysis with Python and pandas'.

Why This Topic Matters

If you are pursuing machine learning or data science using Python, you will inevitably use **pandas**. Raw data is rarely ready for an algorithm; pandas provides a set of tools to **analyze, manipulate, and transform** that data into a format that is usable by machine learning models. Understanding these fundamentals is the first step in your journey to becoming a data scientist or machine learning engineer.

Learning Outcomes

After completing this pre-reading and attending the lecture, you should be able to:

- Define **pandas** and its role in the machine learning workflow.
- Differentiate between the two main pandas datatypes: **Series** and **DataFrames**.
- Import data from external files (like .csv) and export modified datasets.
- Use built-in functions to **describe** and gain a statistical overview of a dataset.
- Select, filter, and manipulate data using **labels** and **integer positions**.
- Address missing values and perform basic **feature engineering**.

Activate Your Prior Knowledge

1. **How do you currently organize data?** Think about using a spreadsheet like Excel or Google Sheets. Consider how you organize columns (features) and rows (entries).
2. **What happens when data is missing?** If you were looking at a list of car sales and some prices were missing, how would that affect your ability to calculate an average?
3. **How do you find a specific "cell" in a table?** Consider whether you find it by its row/column name or by its physical position (e.g., 3rd row, 2nd column).

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Prerequisite Concepts

You are expected to be familiar with basic Python structures, specifically **lists** and **dictionaries**, as these are used to create pandas datatypes from scratch.

Core Concepts

- **The Two Main Datatypes:** A **Series** is a 1-dimensional column of data, while a **DataFrame** is a 2-dimensional table (a collection of Series) with rows and columns.
- **Data Inspection:** Tools like .info() and .describe() allow you to quickly see how many entries exist, whether there are missing values, and provide a statistical summary (mean, max, min) of numerical data.
- **Selection (loc vs. iloc):** Use .loc[] to select data based on **labels** (names) and .iloc[] to select based on **integer position** (index).
- **Data Cleaning:** Real-world data often has missing values (shown as NaN). You can either fill these gaps using .fillna() or remove the incomplete rows with .dropna().
- **Chaining and Inplace:** In pandas, you often add multiple functions together in a row (chaining). Most operations return a copy of the data unless you specifically use the inplace=True parameter or reassign the variable.

Guided Reading Questions

1. **Why is pandas considered an essential tool for machine learning?** Identify the specific use case mentioned regarding how it prepares data for algorithms.
2. **What is the "Anatomy of a DataFrame"?** Look for the main components (Index, Columns, Rows) that make up a standard table.
3. **How does "Boolean Indexing" work?** Observe how you can filter a dataset to show only rows that meet a specific condition (e.g., cars with over 100,000 miles).
4. **Why should you use** .sample() **on large datasets?** Consider the benefits of working with a fraction of a 2-million-row dataset while testing your code.

Reflection Activity

Write a 150–200 word reflection on how you would approach a dataset with **"dirty data"** (e.g., missing values or incorrect datatypes). Based on the reading, explain why you might choose to fill missing values with a mean versus dropping the rows entirely. Mention at least two pandas functions you would use to inspect the data before making your decision.

Self-Assessment

Answer the following before class:

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1. What is the standard abbreviation used when importing pandas?
2. Name the function used to view the **last** 5 rows of a DataFrame.
3. What is the difference between a 1-dimensional and a 2-dimensional pandas datatype?
4. How do you convert a non-numeric column (like a price with a "$" sign) into an integer?
5. What does the .dtypes attribute tell you about your dataset?